Statistical Modelling & Analysis (STAT2004)
Applied Class (Week 4)

We will go through Question 1, 2 & 4 together in the Applied Class. Try the other questions yourself during the week and discuss your results with your tutor next week.

1. The number of trials required to obtain the first success is modelled by the geometric distribution, which has pmf

$$f(x;p) = p(1-p)^{x-1}, \qquad x = 1, 2, 3, \ldots \qquad p \in (0, 1].$$

It can be shown that if $X \sim \mathsf{Geometric}(p)$, then $\mathbb{E}_p(X) = p^{-1}$ and $\mathrm{Var}_p(X) = p^{-2}(1-p)$. Suppose $X_1, \ldots, X_n$ is a random sample from this distribution. Let p_0 denote the true value of the parameter.

(a) Determine the expected information and observed information.

Solution. The likelihood is

$$\mathcal{L}(p) = \prod_{i=1}^{n} p(1-p)^{x_i - 1} = p^n (1-p)^{\sum_{i=1}^n x_i - n}.$$

Hence the log-likelihood is

$$\ell(p) = n \log p + \left(\sum_{i=1}^{n} x_i - n \right) \log(1-p).$$

The score function is

$$U(p) = \ell'(p) = \frac{n}{p} - \frac{\sum_{i=1}^n x_i - n}{1-p}.$$

The second derivative is

$$\ell''(p) = -\frac{n}{p^2} - \frac{\sum_{i=1}^n x_i - n}{(1-p)^2}.$$

Therefore the observed information is

$$J(p) = -\ell''(p) = \frac{n}{p^2} + \frac{\sum_{i=1}^n x_i - n}{(1-p)^2}.$$

Taking expectation under p, and using

$$\mathbb{E}_p \left(\sum_{i=1}^{n} X_i \right) = \frac{n}{p},$$

gives

$$I(p) = \mathbb{E}_p \left(J(p) \right) = \frac{n}{p^2} + \frac{n/p - n}{(1-p)^2} = \frac{n}{p^2(1-p)}.$$

(b) Suppose $p_0 \in (0, 1)$. State the asymptotic distribution of $\sqrt{n}(\hat{p} - p_0)$, where $\hat{p}$ is the MLE.

Solution. From Question 1(c) of the Week 3 Applied Class, the MLE is

$$\hat{p} = \frac{n}{\sum_{i=1}^n X_i} = 1/\bar{X}.$$

For $p_0 \in (0, 1)$, the usual MLE asymptotic normality result states

$$\sqrt{n}(\hat{p} - p_0) \xrightarrow{d} \mathsf{N}\left(0, \frac{1}{i(p_0)} \right),$$

where $i(p_0)$ is the expected information for one observation. Since

$$i(p_0) = \frac{1}{p_0^2(1-p_0)},$$

we have

$$\sqrt{n}(\hat{p} - p_0) \xrightarrow{d} \mathsf{N}\left(0, p_0^2(1-p_0)\right).$$

(c) What is the distribution of $\hat{p}$ when $p_0 = 1$? Explain why this is not described by the usual asymptotic normality result.

Solution. If $p_0 = 1$, then $\mathbb{P}_{p_0}(X_i = 1) = 1$ for every i. Hence, $X_1 = \cdots = X_n = 1$ with probability one. Therefore,

$$\hat{p} = \frac{n}{\sum_{i=1}^n X_i} = 1$$

with probability one. Thus $\hat{p}$ has a degenerate distribution concentrated at 1. The usual regularity conditions for the standard MLE asymptotic normality theorem apply to interior parameter values, not to this case where p_0 is on the boundary of the parameter space.

(d) Suppose you want to estimate the mean of the distribution, which is given by p^{-1}. Does the MLE of p^{-1} achieve the Cramer-Rao lower bound.

Solution. Let $g(p) = 1/p$. By the equivariance property, the MLE of $1/p$ is $1/\hat{p} = \bar{X}$. Since $\mathbb{E}_p(X) = 1/p$, $\bar{X}$ is an unbiased estimator of $1/p$. The variance of this estimator is

$$\mathrm{Var}_p(\bar{X}) = n^{-1}\mathrm{Var}_p(X) = \frac{(1-p)}{np^2}.$$

We have the expected information in part (a). As $g'(p) = -1/p^2$, the Cramer-Rao lower bound is

$$\frac{(g'(p))^2}{I(p)} = \frac{(-1/p^2)^2}{n/p^2(1-p)} = \frac{(1-p)}{np^2}$$

This is exactly $\mathrm{Var}(\bar{X})$. Hence, the MLE of p^{-1} attains the Cramer-Rao lower bound.

2. Suppose $X_1, \ldots, X_n$ is a random sample from a distribution with probability density function

$$f(x;\theta) = \tfrac{1}{2}(1+\theta x), \qquad x \in (-1,1), \qquad \theta \in [-1,1].$$

Let θ_0 denote the true value of the parameter.

(a) Determine the observed information.

Solution. The log-likelihood is

$$\ell(\theta) = -n\log 2 + \sum_{i=1}^n \log(1+\theta x_i).$$

The score function is

$$U(\theta) = \ell'(\theta) = \sum_{i=1}^n \frac{x_i}{1+\theta x_i}.$$

The second derivative is

$$\ell''(\theta) = -\sum_{i=1}^n \frac{x_i^2}{(1+\theta x_i)^2}.$$

Hence the observed information is

$$J(\theta) = -\ell''(\theta) = \sum_{i=1}^n \frac{x_i^2}{(1+\theta x_i)^2}.$$

(b) Suppose $\theta_0 = 0$. Determine the expected information and state the asymptotic distribution of $\sqrt{n}(\hat{\theta} - \theta_0)$, where $\hat{\theta}$ is the MLE of θ.

Solution. The expected information is

$$I(\theta) = \mathbb{E}_\theta\left(J(\theta)\right) = \mathbb{E}_\theta\left(\sum_{i=1}^{n} \frac{X_i^2}{(1+\theta X_i)^2}\right) = n\mathbb{E}_\theta\left(\frac{X^2}{(1+\theta X)^2}\right).$$

When $\theta = 0$, the probability density function of X is $f(x;0) = \frac{1}{2}$, for $x \in (-1,1)$. So

$$I(0) = \frac{1}{2}\int_{-1}^{1} x^2 dx = \frac{1}{3}.$$

The usual asymptotic normality result gives

$$\sqrt{n}(\hat{\theta} - 0) \xrightarrow{d} \mathsf{N}\left(0, \frac{1}{I(0)}\right) = \mathsf{N}(0,3).$$

(c) Suppose $\theta_0 = -1$. Explain why the usual asymptotic normality result cannot hold for $\sqrt{n}(\hat{\theta} - \theta_0)$.

Solution. Since the parameter space is $[-1,1]$, the MLE must satisfy

$$-1 \le \hat{\theta} \le 1.$$

If $\theta_0 = -1$, then

$$\sqrt{n}(\hat{\theta} - \theta_0) = \sqrt{n}(\hat{\theta} + 1) \ge 0$$

with probability one. A non-degenerate normal random variable cannot be nonnegative with probability one. Hence the usual centred asymptotic normality result cannot hold at the boundary point $\theta_0 = -1$.

3. Recall that if $(N_0, N_1, \ldots, N_k)$ has a multinomial distribution with cell probabilities $(\pi_0, \pi_1, \ldots, \pi_k)$ and total count n, then

$$\mathbb{P}(N_0 = n_0, \ldots, N_k = n_k) = \frac{n!}{n_0! n_1! \cdots n_k!}\pi_0^{n_0}\pi_1^{n_1}\cdots\pi_k^{n_k},$$

where $n_0 + n_1 + \cdots + n_k = n$ and $\pi_0 + \pi_1 + \cdots + \pi_k = 1$. We will apply this distribution to the following problem. A certain type of plant cell is classified into four versions. A simple one-parameter model, motivated by genetic considerations, assigns the following probabilities to the four versions:

$$(2+\theta)/4, \quad (1-\theta)/4, \quad (1-\theta)/4, \quad \theta/4,$$

where $\theta \in (0,1)$ is unknown. The biological details are not important for this question; the aim is to estimate θ in this constrained multinomial model. Suppose a sample of n cells has the observed frequencies (n_0, n_1, n_2, n_3), where $n = n_0 + n_1 + n_2 + n_3$.

(a) Show that, up to a multiplicative constant not depending on θ, the likelihood depends on n_1 and n_2 only through $n_1 + n_2$.

Solution. The likelihood is

$$\begin{aligned}\mathcal{L}(\theta) &= \frac{n!}{n_0! n_1! n_2! n_3!}\left(\frac{2+\theta}{4}\right)^{n_0}\left(\frac{1-\theta}{4}\right)^{n_1}\left(\frac{1-\theta}{4}\right)^{n_2}\left(\frac{\theta}{4}\right)^{n_3} \\ &= \frac{n!}{n_0! n_1! n_2! n_3!}\left(\frac{2+\theta}{4}\right)^{n_0}\left(\frac{1-\theta}{4}\right)^{n_1+n_2}\left(\frac{\theta}{4}\right)^{n_3}\end{aligned}$$

Hence the likelihood depends on n_1 and n_2 only through their sum $n_1 + n_2$.

(b) Does the MLE exist when $n_1 + n_2 = 0$? Justify your answer. Identify the necessary and sufficient conditions for the MLE to exist.

Solution. Let $m = n_1 + n_2$. Ignoring constants, the log-likelihood is

$$\ell(\theta) = n_0 \log(2+\theta) + m \log(1-\theta) + n_3 \log \theta, \qquad 0 < \theta < 1.$$

If $m = 0$, then

$$\ell(\theta) = n_0 \log(2+\theta) + n_3 \log \theta,$$

which is increasing in θ on $(0, 1)$. Therefore, its supremum occurs as $\theta \uparrow 1$, but $\theta = 1$ is not in the parameter space. Hence, the MLE does not exist when $n_1 + n_2 = 0$.
More generally,

$$\ell'(\theta) = \frac{n_0}{2+\theta} - \frac{m}{1-\theta} + \frac{n_3}{\theta}$$

and

$$\ell''(\theta) = -\frac{n_0}{(2+\theta)^2} - \frac{m}{(1-\theta)^2} - \frac{n_3}{\theta^2} \leq 0.$$

Hence, $\ell'(\theta)$ is decreasing in θ. If $m > 0$ and $n_3 > 0$, then

$$\ell'(\theta) \to +\infty \quad \text{as } \theta \downarrow 0, \qquad \ell'(\theta) \to -\infty \quad \text{as } \theta \uparrow 1,$$

so the MLE exists (and is unique).
If $m > 0$ and $n_3 = 0$, then $\ell'(\theta) = 0$ has a solution in $(-1, 1)$ if and only if $\ell'(0+) = \frac{n_0}{2} - m > 0$, that is, when $n_0 > 2m$. If $n_0 \leq 2m$, the supremum of the log likelihood is approached as $\theta \downarrow 0$, which is outside the open parameter space.
Therefore, with $m = n_1 + n_2$, the MLE exists if and only if

$$m > 0 \quad \text{and} \quad \big(n_3 > 0 \text{ or } n_0 > 2m\big).$$

(c) Determine an expression for the MLE of θ assuming it exists.

Solution. Let $m = n_1 + n_2$. Setting the score function equal to 0,

$$\frac{n_0}{2+\theta} - \frac{m}{1-\theta} + \frac{n_3}{\theta} = 0.$$

Multiplying by $\theta(1-\theta)(2+\theta)$ and simplifying gives

$$n\theta^2 + (n_3 + 2m - n_0)\theta - 2n_3 = 0.$$

The root in $(0, 1)$, when it exists, is

$$\hat{\theta} = \frac{n_0 - n_3 - 2m + \sqrt{(n_3 + 2m - n_0)^2 + 8nn_3}}{2n}.$$

Hence, replacing m by $n_1 + n_2$,

$$\hat{\theta} = \frac{n_0 - n_3 - 2(n_1 + n_2) + \sqrt{\{n_3 + 2(n_1 + n_2) - n_0\}^2 + 8nn_3}}{2n}.$$

(d) Determine the asymptotic variance of $\hat{\theta}$.

Solution. From part (b), the observed information is

$$J(\theta) = -\ell''(\theta) = \frac{n_0}{(2+\theta)^2} + \frac{n_1 + n_2}{(1-\theta)^2} + \frac{n_3}{\theta^2}.$$

The expected information is

$$\begin{aligned} I(\theta) = \mathbb{E}_\theta\left(J(\theta)\right) &= \mathbb{E}_\theta\left(\frac{N_0}{(2+\theta)^2} + \frac{N_1+N_2}{(1-\theta)^2} + \frac{N_3}{\theta^2}\right) \\ &= \frac{\mathbb{E}_\theta(N_0)}{(2+\theta)^2} + \frac{\mathbb{E}_\theta(N_1)+\mathbb{E}_\theta(N_2)}{(1-\theta)^2} + \frac{\mathbb{E}_\theta(N_3)}{\theta^2} \\ &= \frac{n}{4}\left(\frac{1}{2+\theta} + \frac{2}{1-\theta} + \frac{1}{\theta}\right) \\ &= \frac{n(1+2\theta)}{2\theta(1-\theta)(2+\theta)} = n\,i(\theta). \end{aligned}$$

Therefore

$$\sqrt{n}(\hat{\theta}-\theta_0) \xrightarrow{d} \mathsf{N}\left(0, \frac{1}{i(\theta_0)}\right),$$

so

$$\frac{1}{i(\theta)} = \frac{2\theta(1-\theta)(2+\theta)}{1+2\theta}.$$

Adapted from Azzalini (1999) Statistical Inference - Question 3.19

4. In the Week 2 applied class, we used R to find the maximum likelihood estimator of the zero inflated Poisson distribution. Recall the zero inflated Poisson distribution has probability mass function

$$f(x;\pi,\lambda) = \begin{cases} \pi + (1-\pi)\exp(-\lambda) & x = 0 \\ (1-\pi)\exp(-\lambda)\frac{\lambda^x}{x!} & x = 1, 2, \ldots \end{cases}$$

where $0 \leq \pi \leq 1$ and $\lambda > 0$.

Adapt your code from last week to determine standard errors of the maximum likelihood estimators, and apply your code to the count data in the file `fish.csv`.

Solution. Only minor changes to the code from Week 3 is required. The observed information at the MLE can be obtained from the negative of the Hessian (the matrix of second derivatives) of the log likelihood. This is obtained by setting the argument `hessian` equal to `TRUE` in the `optim` function. The observed information matrix can then be inverted using `solve`.

```
#  The code below is adapted from the Week 3 solution.

fish <- read.csv("https://stats.idre.ucla.edu/stat/data/fish.csv")
data <- fish$count

zip_negloglik <- function(theta, data){
  p <- theta[1]
  lambda <- theta[2]

  loglik <- rep(0, length(data))
  zeros <- (data==0)
  loglik[-zeros] <- log(1-p)  -lambda +
                data[-zeros]*log(lambda) + lfactorial(data[-zeros])
  loglik[zeros] <- log(p + (1-p)*exp(-lambda))
  return(-sum(loglik))
}

# Method of moment estimates

m1 <- mean(data)
```

```
m2 <- mean(data^2)

lambda_hat <- m2/m1 - 1
pi_hat <- 1 - m1/lambda_hat
theta0 <- c(pi_hat, lambda_hat)

# Find the MLE
# We set hessian = TRUE in optim to get the observed information

theta_MLE <- optim(theta0,
                   zip_negloglik,
                   method = "L-BFGS-B",
                   lower = c(1e-8,1e-8),
                   upper = c(1 - 1e-8, Inf),
                   data = data,
                   hessian = TRUE)

theta_MLE$par
[1] 0.567789 7.625909

# Inverting the hessian to get the asymptotic covariance matrix
theta_cov <- solve(theta_MLE$hessian)

# The diagnonal elements of this matrix give estimates of the variance
# of the MLE. Taking the square-root gives the standard error.

sqrt(diag(theta_cov))
[1] 0.03134417 0.26615672
```

5. The inverse Gaussian distribution is often used to model positive, right-skewed waiting times, and it arises as the first passage time distribution of Brownian motion with drift. Let $X_1, \ldots, X_n$ be a random sample from an inverse Gaussian distribution with density

$$f(x; \mu, \lambda) = \left(\frac{\lambda}{2\pi x^3}\right)^{1/2} \exp\left\{-\frac{\lambda(x-\mu)^2}{2\mu^2 x}\right\}, \qquad x > 0,$$

where $\mu > 0$ and $\lambda > 0$. For this parametrisation,

$$\mathbb{E}(X) = \mu, \qquad \mathrm{Var}(X) = \frac{\mu^3}{\lambda}.$$

(a) Determine the score function for $\theta = (\mu, \lambda)$.

Solution. The log-likelihood is

$$\ell(\theta) = \frac{n}{2}\log\lambda - \frac{n}{2}\log(2\pi) - \frac{3}{2}\sum_{i=1}^{n}\log x_i - \frac{\lambda}{2}\sum_{i=1}^{n}\frac{(x_i-\mu)^2}{\mu^2 x_i}.$$

Note that

$$\frac{(x_i-\mu)^2}{\mu^2 x_i} = \frac{x_i}{\mu^2} - \frac{2}{\mu} + \frac{1}{x_i}.$$

Therefore,

$$\frac{\partial}{\partial\mu}\left\{\frac{(x_i-\mu)^2}{\mu^2 x_i}\right\} = -\frac{2x_i}{\mu^3} + \frac{2}{\mu^2}.$$

Hence,

$$U_\mu(\mu,\lambda) = \frac{\partial \ell}{\partial \mu} = \frac{\lambda}{\mu^3}\sum_{i=1}^{n}(x_i-\mu).$$

Also,

$$U_\lambda(\mu,\lambda) = \frac{\partial \ell}{\partial \lambda} = \frac{n}{2\lambda} - \frac{1}{2}\sum_{i=1}^{n}\frac{(x_i-\mu)^2}{\mu^2 x_i}.$$

Thus the score function is

$$U(\mu,\lambda) = \begin{pmatrix} \frac{\lambda}{\mu^3}\sum_{i=1}^{n}(x_i-\mu) \\ \frac{n}{2\lambda} - \frac{1}{2}\sum_{i=1}^{n}\frac{(x_i-\mu)^2}{\mu^2 x_i} \end{pmatrix}.$$

(b) Determine the maximum likelihood estimator of θ.

Solution. Setting $U_\mu(\mu,\lambda)=0$ gives $\sum_{i=1}^{n}(x_i-\mu)=0$, so $\hat{\mu}=\bar{X}$. Setting $U_\lambda(\mu,\lambda)=0$ gives

$$\frac{n}{2\lambda} = \frac{1}{2}\sum_{i=1}^{n}\frac{(x_i-\mu)^2}{\mu^2 x_i}.$$

Therefore, for fixed μ,

$$\hat{\lambda}(\mu) = \frac{n}{\displaystyle\sum_{i=1}^{n}\frac{(x_i-\mu)^2}{\mu^2 x_i}}.$$

Substituting $\hat{\mu}=\bar{x}$ gives the joint MLE

$$\hat{\mu}=\bar{x}, \qquad \hat{\lambda} = \frac{n}{\displaystyle\sum_{i=1}^{n}\frac{(x_i-\bar{x})^2}{\bar{x}^2 x_i}}.$$

Equivalently,

$$\hat{\lambda} = \frac{n}{\displaystyle\sum_{i=1}^{n}\frac{1}{x_i} - \frac{n}{\bar{x}}}.$$

(c) Show that the expected information matrix is

$$I(\theta) = \begin{pmatrix} \frac{n\lambda}{\mu^3} & 0 \\ 0 & \frac{n}{2\lambda^2} \end{pmatrix}.$$

Solution. From part (a),

$$U_\mu(\mu,\lambda) = \frac{\lambda}{\mu^3}\left(\sum_{i=1}^{n}x_i - n\mu\right).$$

Differentiating with respect to μ gives

$$\frac{\partial^2 \ell}{\partial \mu^2} = \lambda\left\{-\frac{3}{\mu^4}\sum_{i=1}^{n}x_i + \frac{2n}{\mu^3}\right\}.$$

Since $\mathbb{E}(X_i)=\mu$,

$$-\mathbb{E}_\theta\left(\frac{\partial^2 \ell(\theta;X)}{\partial \mu^2}\right) = \frac{n\lambda}{\mu^3}.$$

Next,

$$\frac{\partial^2 \ell}{\partial \lambda^2} = -\frac{n}{2\lambda^2},$$

so

$$-\mathbb{E}_\theta \left(\frac{\partial^2 \ell(\theta; X)}{\partial \lambda^2} \right) = \frac{n}{2\lambda^2}.$$

Finally,

$$\frac{\partial^2 \ell}{\partial \lambda \, \partial \mu} = \frac{1}{\mu^3} \sum_{i=1}^{n} (x_i - \mu),$$

and its expectation is

$$-\mathbb{E}_\theta \left(\frac{\partial^2 \ell(\theta; X)}{\partial \lambda \, \partial \mu} \right) = 0.$$

Therefore,

$$I_n(\mu, \lambda) = \begin{pmatrix} \frac{n\lambda}{\mu^3} & 0 \\ 0 & \frac{n}{2\lambda^2} \end{pmatrix}.$$

(d) What does it imply, asymptotically, that the expected information matrix is diagonal?

Solution. The usual asymptotic normality result for the MLE says that

$$\sqrt{n} \begin{pmatrix} \hat{\mu} - \mu \\ \hat{\lambda} - \lambda \end{pmatrix} \xrightarrow{d} N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, i(\theta)^{-1} \right),$$

where $i(\theta)$ is the expected information for one observation. Since the expected information matrix is diagonal, its inverse is also diagonal. Therefore the limiting covariance between $\hat{\mu}$ and $\hat{\lambda}$ is zero.

Thus, asymptotically, the two MLEs are uncorrelated. Because the limiting distribution is joint normal, asymptotic uncorrelatedness also implies asymptotic independence. This does not necessarily mean that $\hat{\mu}$ and $\hat{\lambda}$ are independent in finite samples.